

Hallucination, Formal Reasoning, and the α -Gradient: A Linguistic Framework for AI Architecture

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Abstract

This paper applies the gradient conventionalization framework developed in Mozahid (2025a, 2025b, 2026) to the architecture of artificial intelligence systems, arguing that the persistent divide between statistical and categorical AI is fundamentally a semantic phenomenon rather than an engineering one. The framework operationalizes meaning components through levels of intersubjective agreement (α), providing a principled account of why different AI architectures can process only certain ranges of the α -gradient. Categorical systems operate exclusively at $\alpha \rightarrow 1.0$; large language models (LLMs) operate across $\alpha=0.30\text{--}0.95$, most reliably in the $\alpha=0.70\text{--}0.90$ zone of default entailment. α -level assignments are derived from corpus-distributional measures established independently in Mozahid (2025a), treating AI behavioral data as a convergent test rather than the source of those assignments — a methodological structure that resolves a potential circularity and transforms divergences between predicted and observed AI behavior into theoretically informative evidence. The architectural mapping yields four contributions: it reframes LLM hallucination as *α -level misassignment*, the structural mismatch between assertoric output form and probabilistic generative mechanism; it reframes neuro-symbolic hybrid systems as implicitly *α -stratified architectures* whose routing criterion should be α -level rather than domain or task type; it argues that formal systems are natural language driven to $\alpha \rightarrow 1.0$ through definitional stipulation, connecting the framework to debates in the philosophy of mathematics; and it identifies *α -elevation* — conventionalization's upward pressure on intersubjective agreement — as a theoretical concept currently absent from AI architecture research but necessary for any principled account of how formal reasoning emerges from natural language processing.

Keywords: large language models, hallucination, neuro-symbolic AI, gradient conventionalization, intersubjective agreement, formal reasoning, α -stratification, AI architecture

Hallucination, Formal Reasoning, and the α -Gradient: A Linguistic Framework for AI Architecture

Large language models present a distinctive profile of competence and failure. They handle conversational exchange, literary interpretation, cultural inference, and a wide range of tasks that require sensitivity to contextual, evaluative, and pragmatic meaning with a facility that has surprised even their developers. They fail, often catastrophically, at tasks that require formal precision: mathematical proof, strict logical inference, reliable citation, determinate factual recall. This contrast is well documented but inadequately theorized. The standard accounts — insufficient training data, architectural limitations, the stochasticity of next-token prediction — describe the failure without explaining its structure. They do not tell us why the same system that reasons fluently about the connotations of a word stumbles when asked to verify a simple logical entailment, or why the surface form of a hallucinated output so consistently mimics confident assertion while the underlying process is constitutively uncertain.

This paper proposes a different kind of explanation, one that locates the root of these phenomena not in engineering choices but in the semantic structure of meaning itself. The gradient conventionalization framework, developed across a series of prior papers (Mozahid, 2025a, 2025b, 2026), operationalizes meaning components through levels of intersubjective agreement (α), ranging from 0 — no cross-speaker agreement — to 1 — universal agreement. What the framework reveals, when applied to artificial intelligence systems, is that the divide between statistical and categorical AI maps cleanly onto different regions of this gradient. Statistical systems process content at intermediate α -levels; categorical systems operate at $\alpha \rightarrow 1.0$. This is not a contingent feature of current systems that better engineering might overcome. It reflects the fact that the machinery of statistical learning can only acquire patterns that exist in natural language data, and natural language data, by its nature, occupies the middle

ranges of the α -gradient. The content at $\alpha \rightarrow 1.0$ — the content that formal reasoning requires — is not naturally occurring statistical data. It is the product of a distinct social process: the deliberate act of definitional stipulation through which communities legislate meaning into fixity.

A potential objection to any such mapping must be addressed at the outset. If α -level assignments are derived by observing how AI systems behave — noting that LLMs handle conversational language well and formal reasoning poorly, and then assigning those content types to the α -ranges that explain the pattern — the mapping risks circularity. The explanation would be restating the observation in different vocabulary. This paper forestalls that objection by grounding α -level assignments in corpus-distributional data established independently in Mozahid (2025a), which demonstrates that distributional regularity in large corpora is a reliable proxy for intersubjective agreement. AI behavioral data is then treated as a convergent but independent source of evidence, tested against α -level predictions rather than used to generate them. The theoretical payoff of this structure is considerable: where AI behavior departs from the predictions of the independently grounded mapping, those departures are not anomalies to be explained away but precisely the evidence that reveals most about the relationship between semantic structure and computational mechanism.

This paper develops this argument in seven stages. First, I briefly summarize the gradient conventionalization framework as established in prior work, introducing the concepts and terminology necessary for the application that follows. Second, I describe the statistical-categorical divide in AI as it is currently understood, identifying the limits of existing theoretical accounts. Third, I set out the corpus-based operationalization of α -levels that provides the independent empirical anchor for the architectural mapping. Fourth, I map the α -gradient onto AI architectures, showing that the capacity of a system to process content at a given α -level is determined by its generative mechanism. Fifth, I develop an account of LLM hallucination as α -

level misassignment — a structural mismatch between output form and generative process. Sixth, I reframe neuro-symbolic hybrid systems as implicitly α -stratified architectures, arguing that the routing criterion currently applied heuristically in such systems could be grounded principally in α -measurement. Seventh, I argue that formal systems are continuous with natural language rather than categorically distinct from it, and that this continuity, combined with the dynamic nature of conventionalization as a social process, identifies *α -elevation* as the theoretical concept most critical to any future account of formal reasoning in artificial systems.

The theoretical payoff of this analysis extends in two directions simultaneously. For AI researchers, the framework provides a semantic explanation of empirical phenomena — hallucination, formal reasoning failure, the partial success of hybrid systems — that current architectural accounts do not explain. For linguists and semanticists, the framework's extension to AI systems provides independent confirmation of its core claims about the gradient nature of meaning and the social mechanisms through which fixity is achieved.

The Gradient Conventionalization Framework

The framework developed in Mozahid (2025a, 2026) challenges the traditional distinction in formal semantics between lexical knowledge and encyclopedic knowledge. This distinction, foundational to the field since at least Katz and Fodor (1963), holds that linguistic meaning — systematic, compositional, shared across speakers — constitutes the proper domain of semantic theory, while world knowledge belongs to pragmatics, cultural cognition, or individual psychology. The framework argues that this categorical distinction misrepresents a gradient reality. Both what we call lexical knowledge and what we call encyclopedic knowledge are, at the most fundamental level, learned associations subject to varying degrees of community-wide stabilization. What distinguishes them is not their type but their degree of *conventionalization* —

the extent to which a given meaning component has been stabilized through repeated social coordination among speakers.

This degree of conventionalization is operationalized through *intersubjective agreement*, measured by an agreement score α ranging from 0 to 1. An α -score of 1.0 indicates universal agreement among speakers on the association between a word and a semantic feature; a score of 0 indicates no systematic cross-speaker agreement. Crucially, α is a continuous rather than a categorical variable, and the gradient it defines is populated at all points by natural language meaning.

Several reference levels along this gradient are analytically useful, though they should be understood as convenient points of description rather than natural boundaries. At the highest end (α approaching 1.0), we find *reference*: the basic extensional mapping that determines what entities a word applies to. Virtually all speakers of English agree that *dog* applies to four-legged canine mammals. Slightly lower ($\alpha > 0.90$) lies *denotation*: core definitional features showing very high agreement, such as the inference from *bachelor* to "unmarried adult male." At a moderate-high level ($\alpha = 0.70$ – 0.90), *sense relations* capture how meaning relates to other words in the semantic field. At a moderate level ($\alpha = 0.50$ – 0.70), *cultural connotation* reflects systematic evaluative or stylistic associations — the tendency for *regime* to appear in contexts involving authoritarian governance, for instance — which show community-wide patterning while remaining defeasible and below the threshold of conscious introspective access. Finally, at low levels ($\alpha < 0.50$), *private connotation* reflects purely idiosyncratic associations with minimal cross-speaker agreement.

Correlated with these levels is a typology of entailment strength. *Strong entailments*, characteristic of reference and denotation, are non-defeasible inferences whose denial requires either rejecting the applicability of the word or assigning it a different sense. *Default entailments*,

characteristic of sense relations, are overridable through contextual information but require marked accommodation. *Weak entailments*, characteristic of cultural connotation, are easily cancelled without pragmatic anomaly but nonetheless generate systematic patterns across speakers and corpora. Mozahid (2025a) documents the diagnostic properties of each type in detail, and Mozahid (2026) extends the framework into social epistemology, demonstrating that the gradient also explains the dynamics of disagreement — why scientific disputes tend toward convergence while philosophical and evaluative disputes polarize. Table 1 summarizes the core framework.

Table 1

Conventionalization Levels, Agreement Scores, and Entailment Types

Level	Agreement Score (α)	Entailment Type	Cancellability	Example
Reference	$\rightarrow 1.0$	Strong	Contradictory	<i>dog</i> $\rightarrow$ animal
Denotation	>0.90	Strong	Contradictory	<i>bachelor</i> $\rightarrow$ unmarried
Sense	0.70–0.90	Default	Marked but acceptable	<i>bird</i> $\rightarrow$ flies
Cultural Connotation	0.50–0.70	Weak	Easily overridden	<i>regime</i> $\rightarrow$ oppressive
Private Connotation	<0.50	None	N/A	<i>dog</i> $\rightarrow$ childhood memory

Readers seeking a full derivation of this framework, its empirical support, and its formal properties are referred to Mozahid (2025a, 2026). The present paper takes the framework as established and applies it to a domain — artificial intelligence architecture — that the prior papers did not address.

The Statistical-Categorical Divide in AI

The history of artificial intelligence research can, in one reading, be organized around a persistent divide between two families of approach. On one side stand symbolic and categorical systems: expert systems such as MYCIN and DENDRAL (Shortliffe, 1976; Buchanan & Feigenbaum, 1978), automated theorem provers such as Lean, Coq, and Isabelle (Moura & Ullrich, 2021; Bertot & Castéran, 2004; Nipkow et al., 2002), logic programming systems such as Prolog, ontology reasoners operating over formal knowledge graphs, and planning systems based on formal state-space representations such as PDDL (McDermott et al., 1998). These systems operate through rule application. Their outputs are determined by the application of explicitly specified rules to explicitly specified inputs, and the question of whether an output is correct has a determinate binary answer. On the other side stand connectionist and statistical systems: neural networks of all kinds, and most recently large language models built on the transformer architecture (Vaswani et al., 2017). These systems learn statistical associations from data and generate outputs by sampling from probability distributions over possible next elements in a sequence. Their outputs are not right or wrong in the formal sense but more or less probable, more or less expected given the training distribution.

This divide has been recognized and theorized for decades. The early history of AI was dominated by symbolic approaches, which fell out of favor partly due to the brittleness of hand-specified rule systems and partly due to the difficulty of scaling formal representations to the full complexity of natural language (Dreyfus, 1972; Minsky, 1975). The connectionist revival of the

1980s and the deep learning revolution of the past fifteen years shifted the field toward statistical approaches, which have proven dramatically more scalable. But scalability has come at a cost. Statistical systems handle natural language with increasing fluency while consistently failing at formal reasoning tasks that symbolic systems handle trivially. This has generated a sustained research interest in hybrid or neuro-symbolic approaches (Garcez et al., 2019; Marcus, 2020; Bengio et al., 2021) that attempt to combine the scalability of statistical learning with the precision of formal computation.

The theoretical accounts of this divide in the AI literature are predominantly architectural and computational. The failure of LLMs at formal reasoning is attributed to the stochasticity of autoregressive generation, the absence of a verification step in standard inference, the statistical averaging over training data that conflates correct and incorrect patterns, or more recently to the absence of an explicit search process during inference (Valmeekam et al., 2022; Stechly et al., 2023; Kambhampati, 2024). These are accurate descriptions of mechanisms. What they do not provide is an account of why the divide exists at the level of content — why certain kinds of meaning, and not others, are accessible to statistical processing. The present paper argues that this question requires a semantic answer, and that the gradient conventionalization framework provides one.

Corpus-Based Operationalization of α -Levels

Before the α -gradient can be mapped onto AI architectures in a non-circular way, it must be established that α -levels are measurable independently of the AI behavior they are intended to explain. Mozahid (2025a) demonstrates that intersubjective agreement, while ultimately a property of speaker communities, has reliable corpus-distributional signatures. The methodological foundation of that work makes it possible for the present paper to inherit an empirical operationalization rather than constructing one from scratch.

The core insight is that distributional regularity in large corpora is a proxy for intersubjective agreement. When a term is used by many speakers across many contexts in systematically consistent ways — consistent collocational partners, consistent syntactic environments, consistent pragmatic functions — that consistency is the corpus-level signature of a high α -score. The regularity is not caused by high agreement in some independent sense; it constitutes evidence of it, because corpora are aggregate records of communicative acts across large speaker communities, and systematic patterns in those records reflect the stabilized conventions those communities share.

Several corpus-computable measures operationalize this relationship. *Collocational entropy* measures the distributional diversity of a term's collocational partners: low entropy indicates that the term consistently co-occurs with a narrow, predictable set of items, which corresponds to high α . Mathematical and logical terms such as *if and only if*, *qua*, and *converges* exhibit near-zero collocational entropy across both specialized and general corpora; their distributional environments are almost entirely predictable. Evaluative cultural terms such as *regime*, *militant*, and *radical* exhibit moderate entropy: their collocations are patterned and non-random, but admit considerably more variation than formal terms. Idiosyncratic or private associations leave almost no consistent collocational trace at all, appearing in effectively random distributional environments.

Cross-register stability provides a complementary measure. A term with high α should mean roughly the same thing — should occupy similar distributional environments — across registers as different as legal prose, newspaper reportage, informal conversation, and academic writing. Terms with lower α show register-dependent distributional shifts that reflect the community-specific and context-sensitive nature of their associations. The term *force* in ordinary discourse collocates with physical, social, and metaphorical contexts in roughly equal

proportion; in physics texts its distributional environment narrows sharply to a specific technical domain, reflecting the α -elevation that formal definition has imposed on that specialized community.

Definitional presence in reference corpora — legal codes, mathematical handbooks, technical glossaries, dictionary entries — provides a third independent indicator. Terms that have received explicit definitional treatment in authoritative reference works have, by that very fact, undergone a social act of α -legislation of the kind the framework describes. Their presence in definitional contexts is a corpus-measurable sign that a community has stipulated their meaning to $\alpha \rightarrow 1.0$. Automated theorem provers, legal statutes, and formal ontologies can be treated as definitional corpora whose entries are, by construction, at $\alpha \rightarrow 1.0$.

These three measures converge on a consistent picture of the gradient. Mathematical and logical symbols, formally defined legal terms, and technical terms with authoritative definitional records cluster at the high end, with collocational entropy near zero, cross-register stability near maximum, and high definitional presence. Culturally loaded evaluative terms cluster in the moderate range, with patterned but non-rigid distributional environments. Idiosyncratic personal associations — the kind of meaning that varies from speaker to speaker with no systematic community pattern — leave almost no distributional trace.

The important methodological point is that this clustering can be established entirely from corpus analysis, without reference to how any AI system processes the terms in question. The α -level assignments that follow — including the claim that LLMs operate most reliably at $\alpha=0.70$ – 0.90 and that formal reasoning requires $\alpha \rightarrow 1.0$ — are grounded in this independently established distributional picture. AI behavioral data then enters as a second, convergent source of evidence. Where AI performance tracks the corpus-grounded α -level predictions, that convergence confirms both the predictions and the theoretical account. Where AI performance diverges from

those predictions, the divergence is theoretically informative: it reveals the ways in which the relationship between semantic structure and computational mechanism is more complex than a simple mapping would suggest.

This structure has a further advantage: it transforms the fact that AI behavior is an *incomplete* predictor of α -levels from an awkward residual into a theoretical resource. If AI behavioral data perfectly tracked α -level assignments, one could reasonably suspect that the framework was merely redescribing the data. The divergences — the cases where corpus-grounded α predictions and observed AI behavior come apart — are precisely where the framework earns its explanatory keep, because those divergences require an account that goes beyond the behavioral description itself.

Mapping the α -Gradient onto AI Architectures

The central claim of this paper is that the capacity of an AI system to process content at a given α -level is determined by its generative mechanism, and that different generative mechanisms cover different regions of the α -gradient in a principled and predictable way.

Consider symbolic and categorical systems first. Such systems operate through the application of rules to symbols, where the rules are formally specified and the symbols have determinate, non-negotiable meanings assigned by definition. The most elementary fact about rule-based systems is that they require the meanings of their symbols to be fixed absolutely: if the symbol *bachelor* in a knowledge base can mean different things to different reasoners, the inference rules that operate over it will produce inconsistent outputs, and the system will fail. Formal systems tolerate no gradient meaning; gradience in the inputs propagates through derivation steps and, within a few steps, produces outputs that are semantically unintelligible. This is not a design limitation. It is a logical requirement of rule-based inference.

What this requirement amounts to, in the framework's terms, is that symbolic systems can only operate at $\alpha \rightarrow 1.0$. Every symbol in such a system must have an agreement score approaching 1 — not because all speakers of natural language happen to agree on its meaning, but because the process of formal definition has been deliberately used to legislate α to 1.0. When a mathematician writes "let π denote the ratio of a circle's circumference to its diameter," the act of definition is precisely an act of driving α to 1.0 by stipulation. The symbol π thereafter means exactly that, for every participant in the formal system, with no gradient, no connotation, no contextual modulation. Axioms, definitions, and formal conventions are social acts of *α -legislation*: they take content that might occupy various positions on the gradient in ordinary language and artificially fix its agreement score at the maximum. Symbolic AI systems are the mechanization of reasoning over content that has undergone this process.

The implications are immediate and far-reaching. Symbolic systems cannot generate novel $\alpha \rightarrow 1.0$ content; they can only operate over content whose α has already been legislated to 1.0 by human definition. They excel at deriving the consequences of stipulated definitions and at verifying whether a given derivation is valid. What lies beyond their reach is the lower- α content that constitutes most of natural language — the cultural connotations, contextual implicatures, evaluative associations, and default inferences that make up the texture of ordinary human communication. They are, in a precise sense, blind below a certain α -threshold.

LLMs present the complementary picture. A large language model is trained on natural language data — text produced by human speakers in ordinary communicative contexts. That data, by its nature, is densely populated with content in the $\alpha=0.30$ – 0.95 range. It contains the full texture of cultural connotation, sense relations, default inference, and contextual pragmatics. It contains very little content at $\alpha \rightarrow 1.0$, because the overwhelming majority of natural language production does not occur in formal mathematical or logical register, and even within formal

texts, the formal symbols (π , $\exists$, $\vdash$) appear embedded in natural language commentary that carries its own range of α -levels. The model learns statistical associations over this data distribution, and the generative mechanism it develops — next-token prediction over a probability distribution — is constitutively suited to the α -range of its training data.

This is precisely where the corpus-based operationalization of α -levels does its work. The collocational and distributional properties of the training data are not the same as the α -properties of the content it expresses, but they are reliably correlated with those properties as established in Mozahid (2025a). The statistical regularities that an LLM can acquire from training data are the regularities in those distributional patterns — which means that the α -range the model's generative mechanism is calibrated to is, effectively, the α -range that corpus distributional methods can independently detect. The model learns what can be learned from pattern frequency; $\alpha \rightarrow 1.0$ content is, by the nature of definitional fixity, content whose meaning does not vary across contexts and thus presents no distributional variation for a statistical learner to capture.

The critical point is that LLMs are not merely failing to reach $\alpha \rightarrow 1.0$ due to insufficient training or imperfect architecture. They are processing content through a mechanism — probabilistic sampling from learned distributions — that is intrinsically calibrated to content at $\alpha = 0.50\text{--}0.90$. This is the range where patterns are systematic enough to be statistically learnable but not rigid enough to require formal specification. It is, in the framework's terms, the range of default and weak entailment — the inferential territory where patterns hold most of the time, are overridable by context, and resist the kind of necessary-consequence relation that formal proof requires.

This analysis yields a mapping that is summarized in Table 2.

Table 2*Entailment Types, α -Ranges, and AI Architectures*

Entailment Type	α -Range	AI Architecture	Reliability
Strong (necessary)	>0.90	Symbolic/categorical	Complete within defined system
Default (overridable)	$0.70\text{--}0.90$	LLMs	High (sweet spot)
Weak (easily cancelled)	$0.50\text{--}0.70$	LLMs	Degrading
None (idiosyncratic)	<0.50	Neither	N/A

LLMs are best understood, on this account, as *default-entailment engines*. Their competence is highest in the $\alpha=0.70\text{--}0.90$ zone because this is the zone their training mechanism is most precisely calibrated to: patterns regular enough to be learned from distributional data, variable enough to require probabilistic rather than deterministic representation. Their performance degrades as content approaches the extremes of the gradient in either direction. At the low end ($\alpha < 0.50$), there are not enough statistical regularities for the model to learn systematic patterns — private connotation is, by definition, idiosyncratic, and idiosyncrasy does not compress into distributional statistics. At the high end ($\alpha > 0.90$), the model's probabilistic machinery introduces noise precisely where the content demands zero tolerance for noise: formal derivations require that each step be exactly right, but LLM generation produces each step with some probability of error, and errors in formal derivation, unlike errors in conversational implicature, do not merely degrade fluency but invalidate the entire output.

This account explains why LLM performance on formal reasoning tasks does not improve monotonically with model scale. Increasing the model's parameters and training data improves performance across the $\alpha=0.50\text{--}0.90$ range — more data provides better calibration of

the statistical regularities in that zone. But it does not change the fundamental nature of the generative mechanism, which remains probabilistic sampling. A larger probabilistic sampler is still a probabilistic sampler. The content at $\alpha \rightarrow 1.0$ requires a deterministic mechanism, not a better-calibrated probabilistic one. The gap is qualitative, not quantitative.

The incomplete predictiveness of AI behavioral data mentioned in the previous section is visible here. Recent techniques such as chain-of-thought prompting (Wei et al., 2022) and retrieval-augmented generation (Lewis et al., 2020) improve LLM performance on tasks that partially overlap with the $\alpha \rightarrow 1.0$ range, but in ways that are explicable without revising the architectural mapping. Chain-of-thought prompting decomposes formal reasoning tasks into intermediate steps that each sit more comfortably within the $\alpha=0.70\text{--}0.90$ range, reducing but not eliminating the exposure to the $\alpha \rightarrow 1.0$ zone. Retrieval-augmented generation grounds LLM outputs in retrieved text that itself already occupies the high- α range — the model is not generating $\alpha \rightarrow 1.0$ content through its statistical mechanism but borrowing it from external sources that have already legislated that content to $\alpha \rightarrow 1.0$. These are engineering workarounds that confirm the architectural constraint by working around it rather than resolving it.

Hallucination as α -Level Misassignment

The phenomenon of LLM hallucination — the generation of confidently asserted, formally plausible, factually incorrect content — has attracted substantial theoretical attention in recent years. The standard accounts locate its source variously in the model's tendency to prioritize fluency over accuracy (Ji et al., 2023), in the compression of training data into model weights such that specific facts are not reliably stored (Kandpal et al., 2023), in the absence of a grounding mechanism that connects model outputs to verifiable external states of affairs (Guo et al., 2022), or in the autoregressive generation process which commits to each token without the ability to revise earlier decisions in light of later content (Bang et al., 2023). Each of these

accounts identifies a genuine contributing mechanism. None provides a unified explanation of the phenomenon's most distinctive and puzzling feature: that hallucinated outputs systematically mimic the surface form of high-confidence, high-precision assertion while being generated by a process constitutively incapable of producing such assertion.

The gradient conventionalization framework provides a more precise account. Hallucination, on this analysis, is *α -level misassignment*. It occurs when a model produces output with the surface form characteristic of $\alpha \rightarrow 1.0$ content — assertoric form, absence of hedges, the pragmatic implicature of factual certainty — using generative machinery calibrated to the $\alpha=0.50\text{--}0.90$ range. The mismatch is between the form of the output and the process that generated it.

This requires some unpacking. Natural language has developed conventionalized surface forms that signal different levels of epistemic commitment. Flat assertions ("The capital of France is Paris," "The proof proceeds by induction on n ," "Einstein published the special theory of relativity in 1905") signal, by their form, that the speaker has access to content at a very high α -level — content that any competent member of the community would agree on if they checked. Hedged assertions ("I believe the population is around 3 million," "This might be a consequence of the theory") signal intermediate commitment. This surface formal repertoire is part of what LLMs learn from their training data. They learn that certain kinds of claims take assertoric form, and they reproduce that form when generating content of the relevant type — biographical facts, mathematical statements, citations, logical derivations.

But the surface form and the generative mechanism are dissociated. When an LLM generates an assertoric citation — "*Chomsky* (1965) argued that..." — the form signals $\alpha \rightarrow 1.0$ commitment (this is a verifiable historical fact with determinate truth conditions) while the generative mechanism producing it is probabilistic sampling calibrated to the $\alpha=0.50\text{--}0.90$ range.

The model is not retrieving a stored fact; it is generating a statistically plausible completion of the prompt. The most statistically plausible completion may indeed be a correct citation, in which case the output is accurate. But when it is not — when the most statistically plausible completion given the prompt and the model's learned associations is a citation that does not exist or a date that is wrong — the model generates it with the same assertoric form, the same surface signal of $\alpha \rightarrow 1.0$ commitment, as a correct citation. The hallucination is structurally indistinguishable from accurate assertion because both are products of the same generative mechanism; accuracy is a property of the relationship between the output and the world, not a property the mechanism can track.

The corpus-based operationalization developed in the preceding section allows a further diagnostic observation. The content categories most vulnerable to hallucination — citations, dates, proper names, mathematical results, legal statutes — are precisely the categories that corpus distributional methods identify as $\alpha \rightarrow 1.0$: near-zero collocational entropy, maximal cross-register stability, high definitional presence. Their distributional environments in corpora are highly standardized. An LLM trained on such corpora learns that content of this type takes assertoric form, reproduces that form fluently, and does so with no mechanism for distinguishing cases where its probabilistic process happens to track the correct $\alpha \rightarrow 1.0$ content from cases where it generates a plausible-looking but incorrect completion.

This reframing has a precise diagnostic implication. The standard observation that hallucination is more frequent for obscure or low-frequency information than for common or high-frequency information (Kandpal et al., 2023) is straightforwardly explained: for high-frequency content, there is sufficient statistical regularity in the training data that the most probable completion is indeed correct. For low-frequency content, the statistical signal is weak, and the model's probabilistic sampler generates plausible-sounding outputs with no reliable

anchor in fact. But in both cases, the surface form is assertoric. The model does not know when its probabilistic process is tracking $\alpha \rightarrow 1.0$ content and when it is not, because tracking that distinction requires a mechanism the model does not have.

This also illuminates why mitigation strategies that operate at the level of the training or fine-tuning process — training models to say "I don't know" more often, reinforcing hedged outputs, training on accuracy-labeled data — can reduce hallucination rates but cannot eliminate the structural mismatch. The mismatch is not a learned habit that can be unlearned. It is a consequence of the architecture: a probabilistic generative mechanism applied to a content domain (formal, assertoric claims) that requires a different kind of process. Training interventions shift the distribution of outputs, but they do not change the nature of the generative mechanism. Reinforcement learning from human feedback (Bai et al., 2022) and related fine-tuning approaches are capable of calibrating the surface expression of epistemic commitment — training a model to hedge where it previously asserted — but they are operating at the level of pragmatic form rather than the level of the structural mismatch itself. The mismatch persists underneath the recalibrated surface.

The α -level misassignment account also explains the observation that hallucination is particularly prevalent in domains with high surface regularity but high factual specificity: legal citations, mathematical proofs, scientific references, dates, names. These domains have exactly the surface properties that make them look, to a probabilistic sampler, like they have high α : the form is standardized, the register is formal, the pragmatic conventions of assertion are strongly observed. But they have the content properties that make them genuinely $\alpha \rightarrow 1.0$: the facts either are or are not accurate, with no gradient. A model trained on legal texts will learn the surface form of legal citation with high fidelity. It will generate content in that form fluently and confidently. The form is well-represented in the training data; the factual accuracy of any specific

citation is not a statistical property of the data and cannot be learned from it. High surface regularity, which in most domains correlates with high α , is in these domains decorrelated from the factual specificity that constitutes the actual $\alpha \rightarrow 1.0$ content. That decorrelation is precisely what the probabilistic sampler cannot detect.

Neuro-Symbolic Hybrids as α -Stratified Architecture

The recognition that purely statistical systems fail at formal reasoning and purely symbolic systems fail at natural language has motivated a sustained interest in hybrid approaches that combine components of both types. The neuro-symbolic research program (Garcez et al., 2019; Marcus, 2020) encompasses a range of architectures, from systems that use neural components for perception and symbolic components for reasoning, to systems that use neural networks to guide symbolic search, to systems that use symbolic constraints to guide neural generation. The most architecturally sophisticated current example is AlphaGeometry (Trinh et al., 2024), a system that achieved olympiad-level performance on geometry problems by combining a language model with a formal geometry deduction engine. In AlphaGeometry, the language model generates candidate auxiliary constructions — the creative heuristic step that requires sensitivity to geometric intuition — while the symbolic deduction engine verifies whether the construction yields a valid proof — the formal verification step that requires deterministic inference.

The current theoretical basis for such hybrid designs is largely heuristic and domain-specific. AlphaGeometry works on geometry because the task decomposes naturally into a generation phase (requiring flexible, contextual heuristic reasoning) and a verification phase (requiring formal proof checking), and because the geometry domain has well-developed formal representations. The question of how to construct hybrid systems for other domains, or how to determine in general which components of a task should be handled statistically and which

symbolically, does not have a principled answer in existing AI architecture theory. The choice of which component handles which part of the task is made by system designers drawing on domain knowledge and engineering intuition.

The gradient conventionalization framework provides a principled account of what these hybrid systems are doing and what the handoff criterion between their components should be. AlphaGeometry is, in the framework's terms, an α -stratified architecture. The language model component operates in the lower- α space — the generation of auxiliary constructions involves geometric intuition, heuristic pattern recognition, and culturally-embedded familiarity with geometric problem-solving traditions, all of which occupy the $\alpha=0.60$ – 0.85 range of conventionalized mathematical practice. The symbolic verifier operates at $\alpha \rightarrow 1.0$ — proof checking is a matter of verifying that each step follows by necessity from the previous steps under formal rules, a binary question with a determinate answer. The handoff between the two components is a handoff between α -levels, even though the system designers did not frame it in those terms. The corpus-based operationalization developed earlier confirms this independently: geometric intuition terms and heuristic construction vocabulary exhibit moderate collocational entropy and register variation in mathematical corpora, consistent with the $\alpha=0.60$ – 0.85 assignment; the formal proof vocabulary of deduction engines exhibits near-zero entropy and maximal register stability, consistent with $\alpha \rightarrow 1.0$.

This reframing has a direct architectural implication. If the handoff in hybrid systems is fundamentally an α -level handoff, then the routing criterion — the decision of which component processes which content — should be based on the α -level of the content, not on the domain or the task type. The current practice routes content to components based on what the task requires (generation goes to the neural component, verification goes to the symbolic component) because in geometry, this distinction maps cleanly onto the α -level distinction. But this clean mapping is

specific to the geometry domain. In other domains — legal reasoning, scientific argumentation, natural language inference — the distinction between generation and verification is less clean, and the appropriate routing criterion cannot be inferred from the task structure without an independent assessment of the α -level of the content being processed.

A routing criterion grounded in α -level is both more general and more principled. Content whose α -level exceeds a threshold should be routed to categorical processing; content below the threshold should be handled statistically. This formulation is domain-neutral: it applies to geometry, legal reasoning, medical diagnosis, and mathematical proof without requiring separate domain-specific engineering decisions for each. It also has a clear empirical operationalization, since α -levels are measurable through the corpus-distributional methods described in Mozahid (2025a) and summarized in the preceding section. The threshold itself is an empirical question — at what α -level does statistical processing become sufficiently unreliable that categorical processing is necessary? — rather than an arbitrary engineering choice.

It should be noted that such a criterion is not yet implementable in current systems as a fully automated real-time process, since the corpus measures described can currently be applied to predefined content categories but not dynamically to arbitrary content mid-inference. The contribution here is theoretical: identifying what the routing criterion should be based on, and why. The engineering challenges of implementing this criterion in real time remain as future work, but the theoretical grounding provides direction for that work in a way that purely heuristic approaches do not.

The α -stratification account also clarifies the limits of current hybrid architectures. AlphaGeometry succeeds because the α -level handoff in geometry is stable and predictable: candidate constructions are always in the lower- α range, proof verification is always in the $\alpha \rightarrow 1.0$ range. In domains where the α -level of content is more variable or harder to assess in

advance — where a single reasoning step might move from default-entailment territory to strong-entailment territory mid-derivation — current hybrid systems would face a routing problem they are not equipped to solve. Legal reasoning is a clear example: a legal argument may involve the application of formally defined statutory terms ($\alpha \rightarrow 1.0$) to factual descriptions whose evaluation is contextually sensitive ($\alpha = 0.50\text{--}0.70$), with reference to precedents whose interpretation is itself contested across jurisdictions ($\alpha = 0.60\text{--}0.85$). The routing problem in such a domain is not merely computationally complex but semantically complex: the α -level of the relevant content can only be determined by understanding the argumentative function the content is serving, which is itself a matter that requires interpretation. The framework identifies this as a structural challenge for the hybrid research program and suggests that progress requires not just better neural components or better symbolic verifiers, but a principled method for assessing the α -level of content dynamically during processing.

Formal Systems as Conventionalized Natural Language

The deepest theoretical claim of this paper concerns the relationship between formal systems and natural language. The standard view, implicit in most of the AI and cognitive science literature, treats formal systems — mathematics, logic, formal grammars — as categorically distinct from natural language. Natural language is vague, context-dependent, connotatively rich, and gradient; formal systems are precise, context-independent, denotatively rigid, and categorical. On this view, the divide between statistical and categorical AI reflects a deeper divide between two fundamentally different kinds of representational system.

The gradient conventionalization framework implies a different view. If meaning is fundamentally gradient — if what we call lexical knowledge is just encyclopedic knowledge at very high α -levels — then formal systems are not a different kind of thing from natural language. They are natural language content that has been driven to $\alpha \rightarrow 1.0$ through the deliberate social

process of definitional stipulation. The symbols of mathematics and logic are not semantically distinct from natural language words; they are natural language expressions whose agreement scores have been fixed at the maximum by explicit convention.

This position is not without philosophical precedent, and situating it within the relevant debates clarifies both its scope and its limits. Wittgenstein's later work (1953) challenges the idea that mathematical necessity is grounded in a domain of abstract objects, arguing instead that mathematical propositions derive their necessity from their role in communal practices of rule-following. Brandom's normative pragmatism (1994) extends this orientation into an explicit account of how inferential norms are instituted through social practices of giving and asking for reasons. Lakatos (1976) offers a detailed historical account of how mathematical concepts are progressively refined through the social dynamics of proof and refutation, a process in which the meanings of terms shift as the community negotiates their boundaries. What the gradient conventionalization framework adds to this tradition is not the philosophical insight that formal meaning is socially grounded — that much has been argued before — but a precise operationalization of the social mechanism responsible: the progressive rise of α through repeated coordination, measurable in corpus data, culminating in the act of stipulative definition that drives α to 1.0.

The framework also distinguishes itself from the most common constructivist interpretation of this tradition. The claim is not that mathematical truth is merely a social agreement that could have gone otherwise, or that the necessity of mathematical results is an illusion sustained by institutional authority. The claim is more specific: the *semantic fixity* of formal symbols — the property that makes them suitable inputs for deterministic rule application — is a product of the α -legislation process, not of a mind-independent mathematical realm. Once α has been driven to 1.0, the content is genuinely non-defeasible within the system that legislated

it, which is precisely what necessity means in the framework's terms. The apparent metaphysical puzzle of why mathematics is necessarily true dissolves, on this view, not into arbitrariness but into a precise social fact about how α -legislation produces non-defeasibility.

Consider how a mathematical theory is built. It begins in natural language: mathematicians discuss intuitions, propose definitions, debate whether a proposed definition captures the right phenomenon, and eventually converge on a stipulation that the community accepts. The act of stipulation ("let a group be a set with a binary operation satisfying the following four axioms...") drives the α of the term *group* from wherever it sat in informal mathematical discourse — perhaps $\alpha=0.80$, where most mathematicians had roughly similar intuitions but with some disagreement about edge cases — to $\alpha \rightarrow 1.0$, where the meaning is fixed by definition and deviation constitutes error rather than alternative interpretation. The formal system then operates over this $\alpha \rightarrow 1.0$ content, and because α is uniform across all participants in the system, mechanical rule application is possible: any two reasoners applying the same rules to the same $\alpha \rightarrow 1.0$ inputs will derive the same outputs.

This account has several significant consequences. First, it means that the content at $\alpha \rightarrow 1.0$ is not discovered but constructed. Mathematics does not find meanings in the world that happen to have universal agreement; it creates meanings with universal agreement by stipulation. This is consistent with the corpus evidence: the near-zero collocational entropy of formal mathematical terms is not a natural property of the concepts they denote but a direct consequence of the definitional practices of the relevant communities, as those practices are reflected in the distributional patterns of the corpora those communities produce.

Second, this account implies that the process of creating formal systems is itself a natural language process — it takes place in ordinary communicative contexts, using natural language at intermediate α -levels, through the social mechanisms of conversation, debate, and convention.

Formal systems emerge from natural language; they do not replace it. The language in which mathematicians discuss their intuitions before settling on a definition is operating at $\alpha=0.70$ – 0.90 . The language in which they state the definition operates at $\alpha \rightarrow 1.0$. The language in which they discuss whether the definition is the right one is back at intermediate levels. Formal reasoning is embedded in a surrounding context of natural language reasoning at lower α -levels, and the two are in continuous interaction. This is visible in the corpora: mathematical texts are not uniform in their α -level distribution. The formal notation has near-zero entropy; the surrounding expository prose discussing motivation, intuition, and interpretation has the moderate entropy characteristic of the $\alpha=0.70$ – 0.90 range.

Third — and most directly relevant to AI architecture — this account implies that the statistical-to-categorical transition is not a category jump but a movement along the gradient. There is no principled discontinuity between content at $\alpha=0.89$ and content at $\alpha \rightarrow 1.0$. The difference is quantitative: higher agreement, lower defeasibility, more complete conventionalization. This has the important implication that a system capable of modeling the conventionalization process — the social dynamics through which α rises over time — would be capable, in principle, of traversing the gradient from natural language to formal content, rather than requiring a hard architectural switch between two fundamentally different systems. No current AI architecture has a mechanism that models this process. The framework identifies it as a necessary component of any system that aims to integrate natural language processing with formal reasoning at a principled level rather than through domain-specific engineering.

α -Elevation and the Missing Mechanism

The framework's treatment of conventionalization as a dynamic process — one in which α rises over time through repeated social coordination — points toward a theoretical gap in

current AI research that the framework is, so far as the present analysis can determine, uniquely positioned to identify.

In the prior papers, conventionalization is treated primarily as a synchronic phenomenon: at any given time, a meaning component occupies a position on the α -gradient, and that position determines the entailment type and cancellability properties of the inference. But conventionalization is also a diachronic process. Meaning components move along the gradient over time. A term that begins as a metaphorical extension with $\alpha=0.50$ — loosely understood, variably applied, defeasible in most contexts — may, through repeated use in a community, stabilize into a sense relation with $\alpha=0.80$, and then, through explicit definition in a technical community, become a formal term with $\alpha \rightarrow 1.0$. The history of scientific and technical vocabulary is largely a history of such *α -elevation*: terms that began as ordinary language expressions, underwent progressive conventionalization through disciplinary use, and eventually received formal definitions that fixed their α at the maximum. The word *force* in ordinary language sits at perhaps $\alpha=0.60$; in Newtonian mechanics it has been driven by explicit definition to $\alpha \rightarrow 1.0$. The corpus signature of this history is detectable in diachronic corpora: the collocational entropy of technical terms decreases measurably across historical periods as they undergo progressive conventionalization, and spikes downward sharply at the moment formal definitions enter the literature.

This diachronic dimension of the framework identifies a process that no current AI architecture models. Statistical systems learn the synchronic distribution of content at the α -levels represented in their training data. They do not model the process by which content moves along the gradient. Symbolic systems operate over content that has already been elevated to $\alpha \rightarrow 1.0$; they have no mechanism for generating new $\alpha \rightarrow 1.0$ content, since that requires the social process of stipulative definition which is external to the formal system itself. Neuro-

symbolic hybrids combine the two types of system but still route content to components based on its current α -level; they do not model the elevation process.

The significance of this gap is clearest when we ask what it would take for an AI system to genuinely integrate natural language understanding with formal reasoning, rather than merely routing different content to different components. Such a system would need to be capable of the process by which human communities create formal systems from within natural language — the process of noticing that a given distinction is important, proposing a definition, refining it through discussion, and eventually stipulating a convention that drives α to 1.0. This is the process by which mathematics and logic were developed, and it is a process that is constitutively social: it requires coordination among agents, the negotiation of shared standards, and the mutual recognition of definitional acts as binding. It is, in other words, exactly the process that the framework describes under the heading of conventionalization.

An AI system that modeled this process — that could participate in the social dynamics of α -elevation rather than merely operating within the results of those dynamics — would represent a qualitatively different kind of system from anything currently in existence. It would be capable of generating the $\alpha \rightarrow 1.0$ content that formal reasoning requires, rather than requiring that content to be prespecified by human mathematicians and logicians. Whether such a system is achievable through purely computational means is an open question that the framework does not answer. But the framework is precise about what such a system would need to do: it would need to model the relationship between intersubjective coordination and agreement score, and to simulate the iterative process by which repeated coordination drives α upward.

Several existing research directions bear some relationship to this problem without directly addressing it. Multi-agent AI systems that coordinate toward shared representations (Stone & Veloso, 2000; Foerster et al., 2016) explore the computational dynamics of agent

coordination but do not connect this to the semantics of meaning stabilization. Constitutional AI and related approaches to value alignment (Bai et al., 2022) involve human-AI coordination over normative principles but do not model the process of meaning conventionalization. Work on emergent communication in multi-agent systems (Lazaridou & Baroni, 2020) examines how agents develop shared representational systems but does not connect this to the α -gradient framework or to the distinction between statistical and formal reasoning. Each of these research programs is investigating something related to α -elevation without having the conceptual tool that would allow the connection to be made explicit.

The gradient conventionalization framework provides that tool. It gives a precise characterization of what conventionalization is (the social process of α -elevation), what it produces (content with increasing intersubjective agreement), and what the end state of the process looks like ($\alpha \rightarrow 1.0$ content that supports formal reasoning). Whether this characterization can be operationalized into a computational architecture is an engineering question that lies beyond the scope of this paper. What the framework contributes is the identification of the target: a system that models the dynamics of α -elevation is a system that, in principle, can bridge the gap between statistical and formal reasoning.

Discussion

The analysis developed in the preceding sections has several implications that warrant explicit statement.

The first concerns the framing of the statistical-categorical divide. The dominant framing in AI research treats this divide as an engineering problem: the question is how to bolt symbolic components onto statistical systems, or how to train statistical systems to behave more like symbolic ones, or how to find the right trade-off between the expressiveness of statistical approaches and the precision of symbolic ones. The gradient conventionalization framework

implies that this framing, while practically useful, is theoretically incomplete. The divide is not primarily an engineering problem but a semantic one: it exists because meaning is gradient, and different generative mechanisms can access only certain ranges of that gradient. Engineering solutions that do not address this semantic structure — that treat the two types of system as architecturally incompatible rather than as differently positioned on the α -gradient — will produce systems that are hybrid in their components but not integrated at the level of meaning processing. The corpus-based operationalization developed here provides a means of making this semantic structure visible independently of the AI behavior it is used to explain, which is a precondition for the kind of theoretically grounded engineering the problem requires.

The second implication concerns the concept of hallucination. The α -level misassignment account developed here is more precise than existing accounts in a specific way: it locates the source of hallucination not in any particular engineering choice that could be revised, but in the structural relationship between the generative mechanism and the α -level of the content being generated. This has a diagnostic implication that differs from those of existing accounts. Mitigation strategies that aim to reduce hallucination by training models to express uncertainty more often are targeting the surface form — trying to align the output form with the underlying generative process. This is valuable but it operates at the level of pragmatic calibration, not the level of the mismatch itself. The mismatch will persist as long as the generative mechanism is probabilistic and the content domain is $\alpha \rightarrow 1.0$. A complete solution to hallucination requires either constraining the system to avoid generating $\alpha \rightarrow 1.0$ content through its statistical mechanism (which would severely limit its capabilities), providing a verification layer that operates at $\alpha \rightarrow 1.0$ (which is the hybrid approach), or developing a mechanism that can genuinely operate at $\alpha \rightarrow 1.0$ (which is the α -elevation problem). Current approaches such as retrieval augmentation and chain-of-thought prompting are best understood as pragmatic implementations

of the first two strategies, which is why they reduce hallucination rates without eliminating the structural conditions that produce it.

The third implication is for the neuro-symbolic research program specifically. The α -stratification account predicts that hybrid systems will succeed most reliably in domains where the α -level of content is stable and predictable — where it is clear in advance which content belongs to the statistical component and which to the symbolic component. It predicts that hybrid systems will face the most difficulty in domains where α -levels are variable within a single reasoning chain — where a derivation might pass through content at $\alpha=0.70$, then $\alpha=0.85$, then $\alpha \rightarrow 1.0$, within a single argument. Legal reasoning is a plausible example: legal argument involves the application of formally defined statutes ($\alpha \rightarrow 1.0$) to factual situations described in natural language ($\alpha=0.50\text{--}0.80$) with reference to precedents whose interpretation is itself contested ($\alpha=0.60\text{--}0.85$). The routing problem in such a domain is genuinely difficult, and the α -stratification account suggests that this difficulty is not merely a matter of computational complexity but of the semantic structure of the domain itself.

The fourth implication concerns the philosophy of formal knowledge. The account developed in the preceding section — that formal systems are natural language driven to $\alpha \rightarrow 1.0$ through definitional stipulation, in a tradition that includes Wittgenstein's rule-following considerations, Brandom's normative pragmatism, and Lakatos's account of mathematical concept evolution — has implications beyond AI architecture. If this account is correct, then the apparent distinction between contingent empirical knowledge and necessary formal knowledge reduces, at the level of semantic mechanism, to a difference in α -level: formal knowledge is not a different kind of knowledge but knowledge whose content has been socially stabilized to the point of non-defeasibility through the process of α -legislation. This resolves the puzzle not by denying the necessity of mathematical truth but by explaining how that necessity is produced —

through a social process of conventionalization that the corpus record documents and that the α -gradient framework describes. This is a claim that reaches beyond the scope of the present paper, which is primarily concerned with AI architecture, but the convergence of the framework with longstanding positions in the philosophy of mathematics is a mark of its coherence rather than an artifact of its application.

Conclusion

This paper has argued that the gradient conventionalization framework developed in Mozahid (2025a, 2025b, 2026), applied to the architecture of artificial intelligence systems, provides a semantic account of phenomena that existing architectural theories describe mechanically but do not explain structurally. The central analytical move has been the mapping of AI architectures onto the α -gradient: symbolic systems operate at $\alpha \rightarrow 1.0$, LLMs operate at $\alpha = 0.30\text{--}0.95$ with greatest reliability in the default-entailment zone of $\alpha = 0.70\text{--}0.90$, and the divide between them reflects not a difference in engineering quality but a difference in which region of the gradient each type of mechanism can access. The α -level assignments underlying this mapping are grounded in corpus-distributional measures established in Mozahid (2025a), making AI behavioral data a convergent and independent test of those assignments rather than their source. This methodological structure resolves the circularity that would otherwise undermine the mapping, and transforms the incompleteness of AI behavior as a predictor of α -levels into theoretical evidence rather than a residual problem.

From this mapping, four substantive conclusions follow. LLM hallucination is *α -level misassignment* — the structural mismatch between assertoric output form and probabilistic generative mechanism — and this structural origin means that it cannot be fully resolved through training interventions that operate at the level of output calibration. Neuro-symbolic hybrid systems are implicitly *α -stratified*, and the principled routing criterion for their components is the

α -level of the content being processed, not the domain or task type; this criterion is more general than existing heuristics and has a clear empirical operationalization through the corpus-distributional methods already established in the prior literature. Formal systems are not categorically distinct from natural language but are the product of the social process of α -elevation — definitional stipulation that drives content from intermediate agreement scores to $\alpha \rightarrow 1.0$ — making the statistical-to-categorical transition a movement along a continuum rather than a category jump, a position that connects the framework to the social-pragmatist tradition in the philosophy of mathematics. The concept of *α -elevation* identifies a process that current AI architecture does not model and that constitutes the core unsolved theoretical problem for any system aimed at integrating natural language processing with formal reasoning.

The analysis here is diagnostic and theoretical. It does not propose specific architectural implementations, and it does not evaluate any particular AI system in detail beyond the case of AlphaGeometry. What it establishes is that the existing theoretical vocabulary of AI architecture research is incomplete in a specific, remediable way: it lacks a semantic account of why different systems can process different kinds of content, and it lacks a concept for the dynamic process through which formal content emerges from natural language through social coordination. The gradient conventionalization framework, grounded in independently verifiable corpus-distributional evidence, provides both.

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